**ANSWER KEY**

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · A2 · PLANT DIVERSITY

# Plant groups and adaptations

★☆☆
Grade 4

**Learning goals**

- Sort plants into groups.
- Match adaptations to habitats.

## 1 Sort these living things. [8]

SORT THESE

**mushroom** **fern** **mango tree** **rice** **mould on bread** **lotus** **pine tree** **moss**

| Flowering plant | Non-flowering plant | Not a plant |
|---|---|---|
| rice | fern | mushroom |
| mango tree | moss | mould on bread |
| lotus | pine tree | |

## 2 Match the plant to its adaptation. [5]

Draw a line to match each pair, or write the letter.

| | | |
|---|---|---|
| mangrove ● | → E | ● **A** thick stem stores water; spines instead of leaves |
| water hyacinth ● | → D | ● **B** waxy leaves that water rolls off |
| cactus ● | → A | ● **C** bendy trunk that does not snap in storms |
| lotus ● | → B | ● **D** spongy stalks full of air to float |
| coconut palm ● | → C | ● **E** breathing roots that stick up out of salty mud |

## 3 Which of these living things is not a plant? [1]

**A** 

**B** 

**C** 

**D** 

○ A
● **B**
○ C
○ D

## 4 True or false? [4]

| Statement | True | False | If false, correct it |
|---|---|---|---|
| 1. Ferns grow from spores. | ✔ | | |
| 2. Rice plants do not have flowers. | | ✔ | Rice has small green flowers. |
| 3. Mangrove trees live in salty water. | ✔ | | |
| 4. A cyclone can bring salty water onto farmland. | ✔ | | |

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · A2 · PLANT DIVERSITY

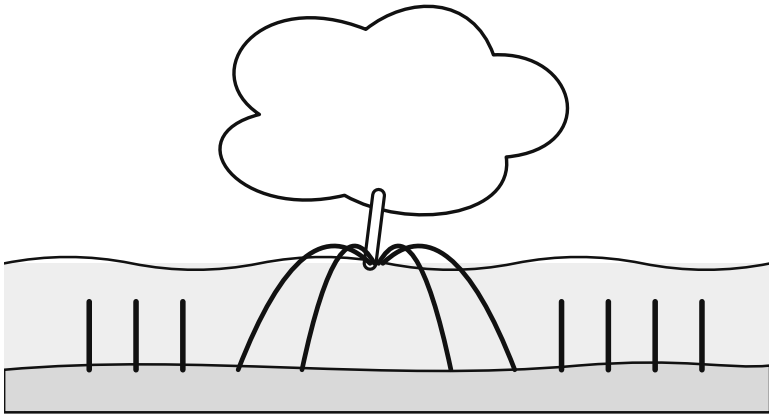
# Why plants look the way they do

★★☆
Grade 4

**Learning goals**

- Explain adaptations.
- Predict the effects of change.

**1** **Look at the mangrove.** [4]



**a)** Why does the mangrove need roots that grow **up** out of the mud?

The wet salty mud has very little air, so the roots grow up into the air to breathe.

**b)** How do prop roots help in a cyclone?

They spread out like legs and hold the tree firmly in the soft mud, so the wind does not knock it over.

**2** **Tisha says, "A cactus would grow well in the Sundarbans." Do you agree? Explain.** [2]

No. A cactus is adapted to dry places; in the wet, salty, muddy Sundarbans its roots would rot and it could not survive.

**3** **A farmer's field near the coast is flooded by salty sea water after a storm. Predict what will happen to the rice and explain.** [2]

The rice will probably die or grow badly, because rice is not adapted to salty water; salt stops the roots taking in water.

**4** **Design a plant that could live on a sunny, windy sand bank in a river (a char). Draw it and label two adaptations.** [3]

My plant

e.g. long roots to reach water and hold the sand; small waxy leaves to save water; bendy stem for wind.

ANSWER KEY

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LEARN · PLAY · WONDER

INVESTIGATE · A2 · PLANT DIVERSITY

# Does a waxy coat stop leaves drying out?

★★☆
Grade 4

**Learning goals**

- Carry out a fair test.
- Measure mass.
- Explain results using adaptation.

Many leaves have a **waxy** coat. Class 4A made model leaves from wet paper towels. They rubbed petroleum jelly (like wax) on some and left others plain. They weighed them, left them in the sun, and weighed them again after 3 hours.

**1** **Complete the plan.** [4]

| | |
|---|---|
| Question | Does a waxy coat stop a leaf losing water? |
| Change | wax on the paper towel or no wax |
| Measure | mass of water lost (g) |
| Keep the same | size of paper towel, amount of water, time, place in the sun |

**2** **Here are the results.** [5]

| Model leaf | Mass at start (g) | Mass after 3 hours (g) | Water lost (g) |
|---|---|---|---|
| plain 1 | 20 | 8 | 12 |
| plain 2 | 20 | 9 | 11 |
| waxy 1 | 20 | 17 | 3 |
| waxy 2 | 20 | 18 | 2 |

**a)** Work out the water lost for each model leaf.

**b)** Which leaves lost less water?

the waxy ones

**3** **Write a conclusion.** [2]

A waxy coat stops leaves losing so much water, so it helps plants survive in dry or sunny places.

**4** **Why did they test two of each kind of model leaf?** [1]

To check the results; if both are similar the results are more reliable.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · A2 · PLANT DIVERSITY

# Water hyacinth: a problem or a gift?

★★★
Grade 4

**Learning goals**

- Weigh up good and bad effects.
- Link science to jobs and design.
- Give a reasoned opinion.

Science · Economics · Design · Environment

**Water hyacinth** (*kochuripana*) came from South America more than a hundred years ago. It grows very fast in warm, still water. It can double in size in about two weeks, covering ponds, canals and rivers like a green carpet.

This causes problems. Boats get stuck, fish have less light and air, and mosquitoes can breed in the still water underneath. But people in Bangladesh have found clever uses for it. Farmers pile it up to make **floating gardens** (*dhap*) to grow vegetables during floods. It is used as **compost** and cattle food, and craft workers weave the dried stems into baskets, mats and furniture that are sold around the world.

Based on information about water hyacinth and floating gardens in Bangladesh.

**1** **Sort the effects.** [6]

SORT THESE

**baskets and mats** **floating gardens** **blocks boats** **mosquitoes breed** **compost for fields** **less light for fish**

| Good for people | Bad for people or nature |
|---|---|
| floating gardens | blocks boats |
| baskets and mats | less light for fish |
| compost for fields | mosquitoes breed |

**2** **Which adaptation helps water hyacinth to cover a pond so fast?** [2]

It floats on spongy stalks full of air and grows and spreads very quickly in warm water.

**3** **Maths. A patch of water hyacinth covers 1 square metre. It doubles every 2 weeks. How much will it cover after 8 weeks?** [2]

| Weeks | 0 | 2 | 4 | 6 | 8 |
|---|---|---|---|---|---|
| Area (square metres) | 1 | 2 | 4 | 8 | 16 |

**4** **Your opinion: Is water hyacinth a problem or a gift? Give reasons for both sides, then decide.** [3]

It is a problem because it blocks rivers and harms fish, but a gift because people make gardens, compost and crafts from it. Best answer: it is useful if people harvest it and stop it spreading.

**ANSWER KEY**

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · A2 · PLANT DIVERSITY

# Show what you know

★★★
Grade 4

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

## 1 Find the key words.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| F | L | W | F | H | A | D | A | P | T | R |
| G | L | F | E | R | N | A | A | M | T | M |
| F | J | M | O | R | W | Y | J | O | E | I |
| L | R | A | E | Y | S | R | O | O | T | S |
| O | D | N | C | B | C | F | A | I | A | I |
| W | S | G | O | E | V | E | T | K | V | C |
| E | A | R | R | O | L | S | E | E | D | A |
| R | D | O | T | F | H | J | B | K | S | C |
| M | F | V | I | N | A | M | O | S | S | T |
| H | F | E | K | H | I | Y | M | H | W | U |
| M | H | U | L | S | P | O | R | E | E | S |

**adapt cactus fern flower mangrove moss roots seed spore**

## 2 Which plant makes spores instead of seeds? [1]

- ○ rice
- ● **fern**
- ○ mango tree
- ○ lotus

## 3 Why does water hyacinth float? [1]

- ○ its roots are heavy
- ● **its stalks are full of air**
- ○ it has no leaves
- ○ it is a fungus

## 4 Draw and label a plant from a dry place and a plant from a wet place. Show one adaptation on each. [4]

dry place
cactus: thick stem stores water, spines

wet place
lotus: waxy floating leaves or mangrove: breathing roots

**5** **Challenge. The sea level is rising and more salty water is reaching farms in the south of Bangladesh. How could scientists help farmers?** [2]

They can breed rice that can grow in salty water (salt-tolerant rice), suggest crops that like salt, or build embankments to keep the sea water out.